AM-I

SOMAIYA VIDYAVIHAR UNIVERSITY

Maximum Marks: 30	**Semester: Aug 2025 – Nov 2025** **Examination: Mid-Semester Examination** **MSE**	**Duration:** 1 Hr 15 Min
Programme code: **Programme: B.Tech**	**Class:** FY	**Semester:** I (SVU R-2025)
Name of the Constituent College: **K. J. Somaiya School of Engineering**	**Name of the department:** COMP/ETRX/EXTC/IT/MECH	
Course Code: 316U06C101	**Name of the Course: Applied Mathematics-I**	

Question No.		Max. Marks
Q:1	Reduce the matrix $A = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 2 & 2 \\ 7 & 4 & 10 \\ 8 & 5 & 8 \end{bmatrix}$ to Normal form and hence find the rank	06
Q:2	**Attempt any TWO from the following.**	12
1	If $u = x(1-y)$, $v = xy(1-z)$, $w = xyz$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$	06
2	Encrypt and decrypt the message "I AM FINE " using the orthogonal matrix $Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	06
3	If $u = x^2y + y^2z + z^2x$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 2(x+y+z)$	06
Q:3	**Attempt any TWO from the following.**	12
1	Use Gauss-Seidel method to solve the following equations (Take three iterations) $15x + y + z = 17$ $2x + 15y + z = 18$ $x + 2y + 15z = 18$	06
2	For which values of λ following set of equations is consistent? Find and solve equations for those values $x + 2y + z = 3,\ x + y + z = \lambda,\ 3x + y + 3z = \lambda^2$	06
3	If $u = x^3 \sin^{-1}\left(\frac{\sqrt{y}+\sqrt{x}}{\sqrt{y}-\sqrt{x}}\right)$ then prove that i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3u$ ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 6u$	06

EP

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: August 2025 – December 2025		
Maximum Marks: 30 **Examination: Mid-Semester Examination**		**Duration: 1 Hr 15 Min**
Programme code: 06 **Programme: B.Tech**	**Class: FY**	**Semester: I** **(SVU-2025)**
Name of the Constituent College: **K. J. Somaiya School of Engineering**	**Name of the department/Branch:** All	
Course Code: 316U06C102	**Name of the Course: Engineering Physics**	

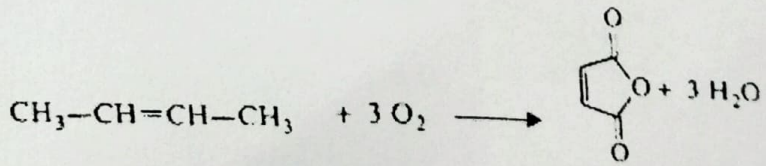
Instructions: **(a)** Draw neat diagram wherever required.
(b) Assume suitable data and symbols if required.

Useful Data: Charge on electron (e) = 1.6×10^{-19} C, Mass of electron (m_e) = 9.1×10^{-31}kg
Avogadro's number (N_0) = 6.023×10^{23}/mole, Planck's constant (h) = 6.63×10^{-34}Js
Boltzmann's constant (k) = 8.625×10^{-5}eV/K=1.38×10^{-23} J/K , 1eV=1.6×10^{-19}J
Permittivity of air (ε_0) = 8.85×10^{-12} F/m, Permeability of air (μ_0) = $4\pi\times10^{-7}$ Wb/Am.

Question No.		Max. Marks
Q1	Attempt any TWO (5 marks each) **a.** With the help of relevant necessary diagram. derive an expression for effective optical path difference for light reflected from thin film of uniform thickness. **b.** What are the different pumping schemes used for laser operation? Give advantages of four level pumping over three level pumping. **c.** What is threshold condition related to LASER ? Derive an expression for amplification factor at threshold condition.	10
Q2	Attempt any TWO (5 marks each) **a.** With the help of proper schematic diagram and necessary steps, derive an expression for numerical aperture of step index optical fiber. **b.** Explain formation of colours in thin film. Why colours which are present in the reflected light are absent in the transmitted light in case of interference in thin film. **c.** What is resolving power of a grating. Explain Rayleigh's criteria for resolution with proper diagram.	10
Q3	Attempt any TWO (5 marks each) **a.** White light falls normally on the soap film (μ=1.34) of thickness 3900 Å. Which wavelength/s within the visible spectrum (4000 Å to 7000 Å) will be intensified in the reflected light? **b.** Find the ratio of population of two energy states of an active medium for radiation of wavelength 694.3nm emitted at room temperature (27°C). Comment on the result. **c.** In a plane transmission grating, the angle of diffraction for second order maxima for a wavelength of 6000 Å is 30^0. Determine the number of lines/cm on the grating surface.	10

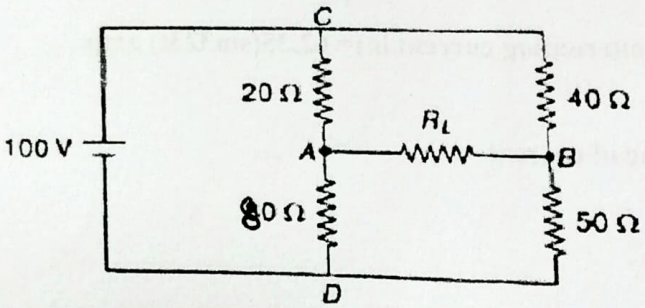
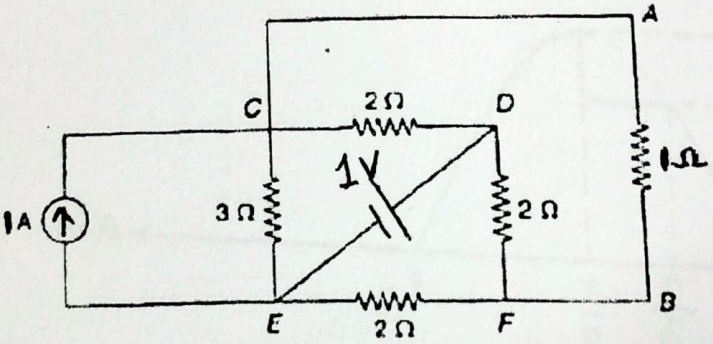
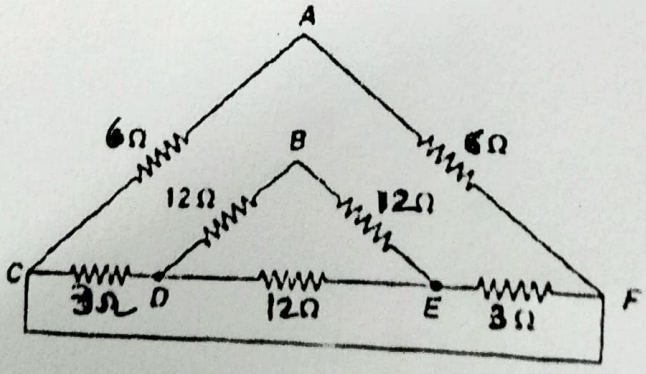
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Semester: Aug 2025 – Nov 20225		
Maximum Marks: 30 **Examination: Mid- Semester Examination** **Duration: 1 Hr 15 Min**		
Programme code: **Programme: UG**	**Class: FY**	**Semester: I (SVU R-2025)**
Name of the Constituent College: **K. J. Somaiya School of Engineering**	**Name of the department:** COMP/ETRX/EXTC/IT/MECH	
Course Code: 316U06C103	**Name of the Course: Engineering Chemistry**	
Information: Ca= 40; Cl= 35.5; H= 1; C= 12; O= 16; S= 32; Si= 28; Mg= 24; Na= 23.		

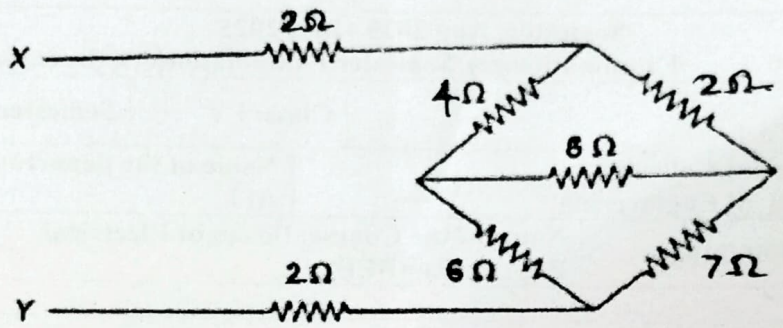
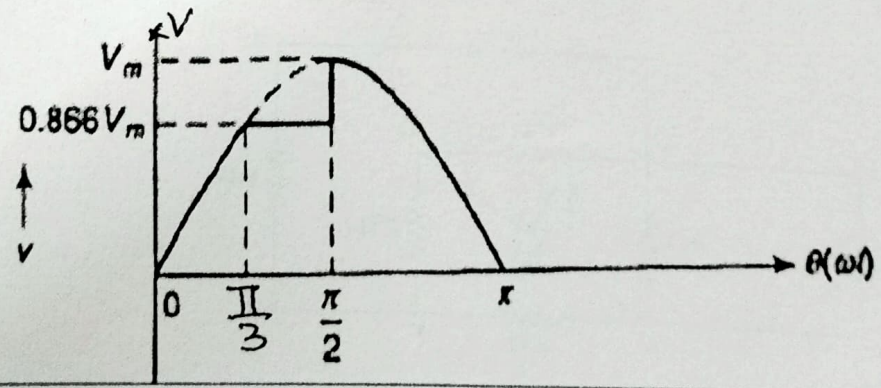
Question No.	Question	Max. Marks
Q1A	Calculate the amount of lime (85%) pure and soda (90%) pure required to soften 10,000 Lit of water containing- $Ca(HCO_3)_2$= 16.2 mg/L; $CaSO_4$= 6.8 mg/L; $CaCl_2$= 11.1 mg/L; SiO_2= 8.0 mg/L; $MgSO_4$= 6.0 mg/L; NaCl= 14 mg/L; $Mg(HCO_3)_2$= 8.4 mg/L. **OR** 20 ml of standard hard water containing 15 g/L calcium carbonate required 25 ml of EDTA for titration. 100 ml of sample hard water required 18 ml of same EDTA. The same sample after boiling required 12 ml of EDTA for titration. Calculate the temporary hardness.	5
Q1B	Differentiate between temporary and permanent hardness.	5
Q2A	Discuss the cold soda-lime process for softening of water in detail. **OR** Discuss the zeolite method of softening of water in detail.	5
Q2B	The hardness of 3500 Lit of water was removed by zeolite softener. The zeolite required 25 Lit of NaCl solution containing 100 gm/Lit of NaCl for regeneration. Calculate the harness of water.	5
Q3A	Explain the principle of 'Atom Economy' with the help of suitable example. **OR** Calculate the atom economy for- 1)  Maleic anhydride 2) $C_6H_5NH_2 + (CH_3CO)_2O \rightarrow C_6H_5NHCOCH_3 + CH_3COOH$ Acetanilide	5
Q3B	Illustrate the traditional and green way to synthesize adipic acid. State the principle involved.	5

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Maximum Marks: 30	Examination: Mid Semester Examination (MSE)	Duration: 1 Hr 15 Min
Programme code: Programme: FY B.Tech	Class: FY	Semester: I (SVU 2025)
Name of the Constituent College: K. J. Somaiya School of Engineering	Name of the department: ALL	
Course Code: 316U06C104	Name of the Course: Basics of Electrical Engineering(BEE)	

Question No.		Max. Marks
Q1	Find the current flowing through R_L in the network using Thevenins theorem. R_L= 10Ω  **OR** Determine current in 1Ω resistor using superposition theorem.  	10M 10M
Q2	A] Find the equivalent resistance between A and B  	5M

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Q2	OR A] Find the equivalent resistance between X and Y.  B] i) Explain Source transformation rules in dc analysis using Suitable example. ii) State and explain Maximum Power transfer theorem.	5M 2 M 3M
Q3	A] The equation of an alternating current is i= 62.35(sin323t) amp. Determine i) Maximum value of current ii) Frequency iii)RMS value iv)Average Value v) Form factor	 1M 1M 1M 1M 1M
Q3	B] Find the RMS value of given waveform.  	5M